

Bot-IoT NetFlow v2 Benchmark Datasheet

Dataset Provenance, Feature Architecture, and Edge Detection Specifications

DISAL Research Project, Bempong and Brown (2026)

1. Benchmark Provenance and Ingestion

The Bot-IoT NetFlow v2 benchmark represents a standardized, flow-aggregated intrusion detection dataset developed by UNSW Canberra Cyber (Sarhan et al., 2021). The raw captures were generated in a simulated Internet of Things network environment incorporating smart home devices, cloud gateways, and node sensors under diverse cyberattacks, including volumetric distributed denial of service (DDoS), denial of service (DoS), reconnaissance, and data theft.

1.1 Partitioning and Class Imbalance

The experimental partition standardizes on a 400,000-sample slice matching the formal benchmark protocol:

- **Total Flow Records:** 400,000 records partitioned into 70% training (280,000), 10% validation (40,000), and 20% test (80,000; exactly 79,889 valid evaluated records).
- **Extreme Class Imbalance:** Severe 278:1 attack-to-normal ratio in the raw partition (99.64% attack traffic vs. 0.36% benign traffic). The test split contains 79,600 attack samples and only 289 normal samples.
- **Baseline Collapse Context:** Standard from-scratch gradient descent collapses to the majority class under this skew, resulting in 0.0% normal traffic recognition (100% false alarm rate). The Transfer Rescue model resolves this by transferring spatial representations from CICIoT2023, restoring normal recognition to 59.5% while maintaining 99.98% attack recall.

1.2 Zero-Leakage Preprocessing Protocol

- **Identity Leakage Prevention:** Raw network IP addresses (`IPV4_SRC_ADDR` and `IPV4_DST_ADDR`) are removed prior to splitting. This prevents deep learning models from memorizing specific host identity subnets rather than learning generalized flow behavior.
- **Strict Training-Set Scaling:** Outlier-robust scaling is executed using a two-stage pipeline: `RobustScaler` followed by `MinMaxScaler`. Both scalers are fitted strictly on the training partition (X_{train}) and applied via transform-only to validation and test sets.
- **Precision Downcasting:** All numerical feature values are cast to 32-bit floats (`float32`) and target labels to 8-bit integers (`int8`) to prevent heap memory exhaustion and optimize CPU execution on edge devices.

2. The 41 NetFlow Feature Specifications

The model consumes 41 standardized bidirectional flow features extracted from network packet exchanges. The catalog below provides formal NetFlow v2 definitions and intrusion detection diagnostic significance.

Feature Name	NetFlow v2 Definition	Intrusion Detection Diagnostic Value
Group 1: Transport and Application Layer Identifiers		
L4_SRC_PORT	Transport layer source port number	Ephemeral client port range versus fixed scanning source
L4_DST_PORT	Transport layer destination service port	Targeted service identifier (e.g., port 80 HTTP, 53 DNS)
PROTOCOL	Internet Protocol suite number	6 for TCP, 17 for UDP, 1 for ICMP message control
L7_PROTO	Detected application layer protocol code	Categorizes application service protocol running on top of flow
Group 2: Traffic Volume and Packet Counts		
IN_BYTES	Incoming volume of bytes sent from source	Volumetric flood magnitude from attacker to victim
IN_PKTS	Incoming packet count from source to destination	Packet burst density indicative of SYN or UDP flooding
OUT_BYTES	Outgoing volume of bytes returned by responder	Distinguishes bidirectional exchanges from dropped packets
OUT_PKTS	Outgoing packet count returned by destination	Zero or low counts signal unacknowledged spoofed requests
Group 3: Cumulative TCP Control Flags		
TCP_FLAGS	Bitwise OR of all TCP flags observed in flow	Overall connection state signature across conversation
CLIENT_TCP_FLAGS	Cumulative TCP flags asserted by flow initiator	Detects client-driven SYN scans, stealth FIN, or RST sweeps
SERVER_TCP_FLAGS	Cumulative TCP flags asserted by responder	Identifies SYN-ACK responses, server resets, or silent drops
Group 4: Flow Timing and Durations		
FLOW_DURATION_MILLISECONDS	Total active conversation duration (ms)	Ultra-short flows denote port scans; long flows denote C2 links
DURATION_IN	Active client transmission duration (ms)	Transmission duration for incoming request payload
DURATION_OUT	Active server response duration (ms)	Response latency and completion profile of remote service

Feature Name	NetFlow v2 Definition	Intrusion Detection Diagnostic Value
Group 5: Packet Length and Hop Limit (TTL)		
MIN_TTL	Minimum Time to Live recorded in flow	Hop count distance to source; identifies spoofed routing
MAX_TTL	Maximum Time to Live recorded in flow	Variance between min and max TTL indicates source hopping
LONGEST_FLOW_PKT	Maximum packet length observed in bytes	Detects oversized payloads used in buffer exploitation
SHORTEST_FLOW_PKT	Minimum packet length observed in bytes	Tiny packet dominance signifies control flooding or ping sweeps
MIN_IP_PKT_LEN	Minimum IP packet header length	Header anomalies, malformed packets, and option padding
MAX_IP_PKT_LEN	Maximum IP packet header length	Identifies maximum transmission unit (MTU) saturation
Group 6: Rate and Throughput Dynamics		
SRC_TO_DST_SECOND_BYTES	Forward transfer speed (bytes per second)	Instantaneous upload velocity from source entity
DST_TO_SRC_SECOND_BYTES	Reverse transfer speed (bytes per second)	Instantaneous download velocity returned by server
SRC_TO_DST_AVG_THROUGHPUT	Average forward throughput (bits per second)	Distinguishes high-speed volumetric floods from trickle scans
DST_TO_SRC_AVG_THROUGHPUT	Average reverse throughput (bits per second)	Reflects server data extraction or service denial saturation
Group 7: Retransmission Reliability Metrics		
RETRANSMITTED_IN_BYTES	Incoming retransmitted byte volume	High volume indicates severe network congestion or SYN drops
RETRANSMITTED_IN_PKTS	Incoming retransmitted packet count	Packet loss rate caused by saturated target buffers
RETRANSMITTED_OUT_BYTES	Outgoing retransmitted byte volume	Unanswered server retries caused by victim exhaustion
RETRANSMITTED_OUT_PKTS	Outgoing retransmitted packet count	Quantifies downstream packet loss on return path
Group 8: Packet Size Distribution Bins		
NUM_PKTS_UP_TO_128_BYTES	Packet count in length bin ≤ 128 bytes	High ratio denotes TCP handshake scans or ICMP probes
NUM_PKTS_128_TO_256_BYTES	Packet count in length bin 129 to 256 bytes	Small message protocols and command-and-control beacons
NUM_PKTS_256_TO_512_BYTES	Packet count in length bin 257 to 512 bytes	Medium-size queries and encrypted TLS client handshakes
NUM_PKTS_512_TO_1024_BYTES	Packet count in length bin 513 to 1024 bytes	Intermediate data payloads and DNS response amplification
NUM_PKTS_1024_TO_1514_BYTES	Packet count approaching standard MTU limit	Full-frame data streaming, large downloads, or exfiltration

Feature Name	NetFlow v2 Definition	Intrusion Detection Diagnostic Value
Group 9: Window and Buffer Dynamics		
TCP_WIN_MAX_IN	Maximum advertised client receive window	Advertised buffer capacity of the connection initiator
TCP_WIN_MAX_OUT	Maximum advertised server receive window	Zero or decaying window advertisements signal target exhaustion
Group 10: Application and Signaling Protocols		
ICMP_TYPE	Control message type field (RFC 792)	8 for echo request, 0 for reply, 3 for unreachable
ICMP_IPV4_TYPE	Sub-classification code for IPv4 ICMP	Identifies port unreachable, fragmentation required, or redirect
DNS_QUERY_ID	DNS transaction identifier number	Zero denotes non-DNS flows; random IDs denote normal queries
DNS_QUERY_TYPE	DNS record type code requested	1 for IPv4 A, 28 for AAAA, 15 for MX, 255 for ANY amplification
DNS_TTL_ANSWER	Cached time to live returned in DNS response	Short TTLs denote fast-flux malicious botnet infrastructure
FTP_COMMAND_RET_CODE	3-digit numeric FTP server return code	220 service ready, 230 login successful, 530 access denied

3. Edge Detector Architectural Profile

The deployed detector couples a hierarchical 1D Swin Transformer backbone with an LSTM temporal sequence classifier to deliver real-time verdicts on resource-constrained hardware.

Metric / Parameter	Value	Operational Significance
Total Model Parameters	1,038,383	Exact match to Bempong and Brown (2026) paper target
Trainable Parameters (TL)	79,301	Fine-tuned temporal head (92.3% parameter freeze ratio)
Full Precision (FP32) Disk Size	4.03 MB	Standard PyTorch weights state dictionary
Quantized (INT8) Disk Size	1.16 MB	3.47x compression ratio for resource-constrained gateways
Attack Detection Recall	99.98%	Only 14 missed attacks out of 79,600 test samples
Overall Accuracy	99.84%	Validated on Bot-IoT NetFlow v2 test split
Inference Latency (INT8)	$\approx$ 85 ms	32-record batch execution on edge CPU hardware
Zero Gradient Overhead	Verified	Evaluation mode with torch.no_grad()

Citation: Bempong, Y., and Brown, R. (2026). *A Viable Hybrid Swin-LSTM Intrusion Detection Architecture for Resource-Constrained IoT Edge Gateways*. DISAL Final Year Engineering Project.